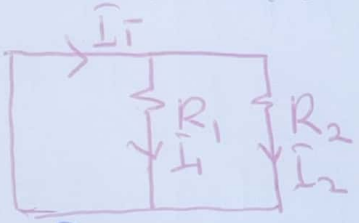
QUESTION 1

- Find total resistance (R_T)
- Find total current ($I_T = \frac{V}{R_T}$)
- Share total current in all branches to find the current in $I_{2\Omega}$
- Use power formulae

① $P = \frac{V^2}{R}$
② $P = I^2R$
③ $P = IV$

> Any can be used but use $P = I^2R$

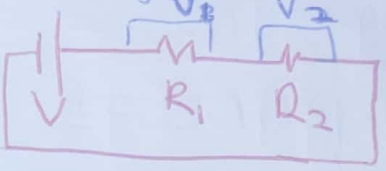
NB Current division rule



$$I_1 = \frac{R_2}{R_1 + R_2} \times I_T$$

$$I_2 = \frac{R_1}{R_1 + R_2} \times I_T$$

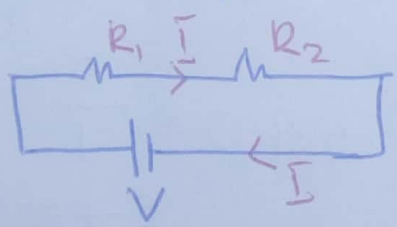
Voltage Division rule.



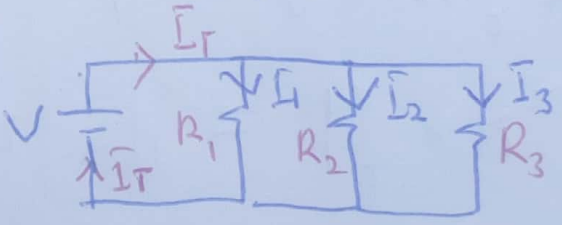
$$V_1 = \frac{R_1}{R_2 + R_1} \times V$$

$$V_2 = \frac{R_2}{R_1 + R_2} \times V$$

NB Series & Parallel combination of resistors.



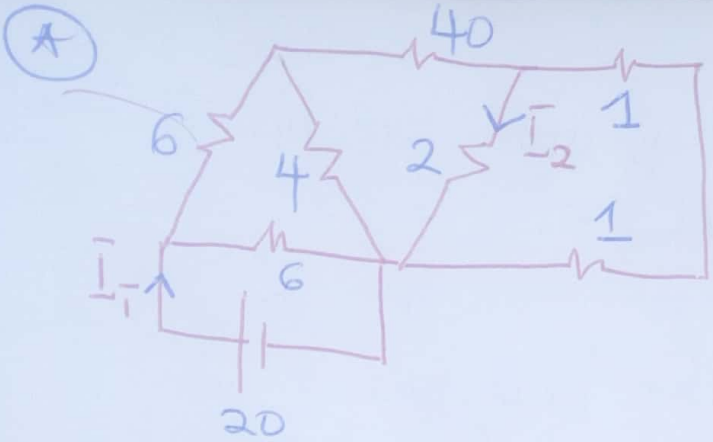
- Same current flows through the resistors



- Different currents flows through the resistors -

NB - I_T will divides between R_1 and the combination of $[R_2 // R_3]$

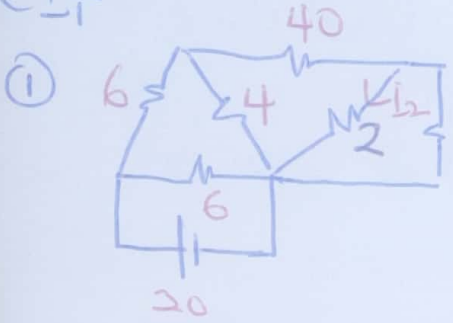
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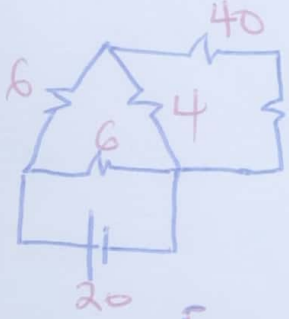
NB

Power $= I_2^2 \times R_2$ (?) (2Ω)

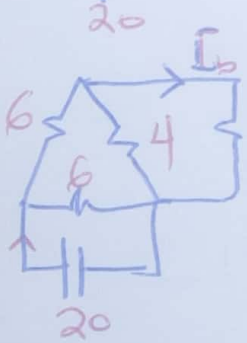
Begin by finding total resistance (I_T)



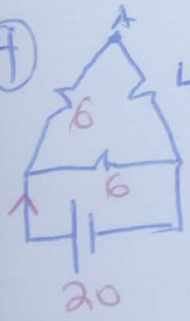
① 1+1 = 2.
1 series 1



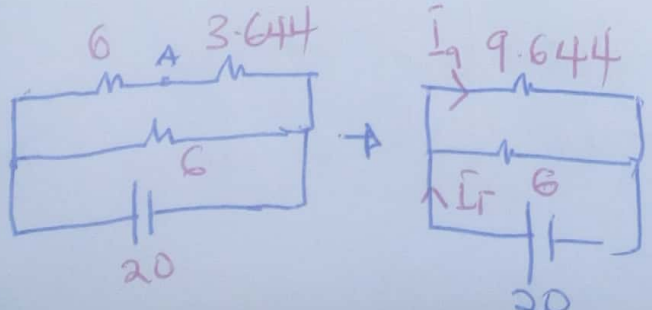
② 2 // 2 = 1
2 parallel 2



③ 40+1 = 41
40 series 1



④ 41 // 41 = 3.644Ω
4 parallel 41



⑤

R_T = total Resistance
$= 9.644 + 1 // 6 = \underline{3.699\Omega}$

I_T from ⑤ $I_T = \frac{V}{R_T} = \frac{20}{3.699}$

$= 5.407\,A$

- Now share 5.407 A through the branches till we get to 2Ω resistor.

From figure ⑤ using Current Division.

$I_q = \frac{20}{9.644}$ or.

$I_q = \frac{6}{6+9.644} \times 5.407\,A.$

$= 2.0737\,A$ //

From figure ③ I_q will be share between 4Ω and 41Ω resistors. (I_b passes through 41Ω which we are interested in.

$I_b = \frac{4}{4+41} \times 2.0737 = 0.1848A$

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Now from figure (1)
Ib will be share between 2Ω and (1+1) ohm's resistors.

$$I_2 = \frac{2}{2+2} \times 0.1843A.$$

$$= \frac{1}{2} \times 0.1843\,A = 0.092$$

NB When dividing current between two equal resistors. divide the current by 2.

$$I_2 = \frac{0.1843}{2} = \boxed{0.09216\,A}$$

(3)

Now we have the current, Hence we can find the power via the two ohm's resistors

$$P_2 = I_2^2 \times R_2.$$

$$P_2 = 0.09216^2 \times 2$$

$$= 0.016\,99\,W$$

$$= \boxed{0.017W}$$

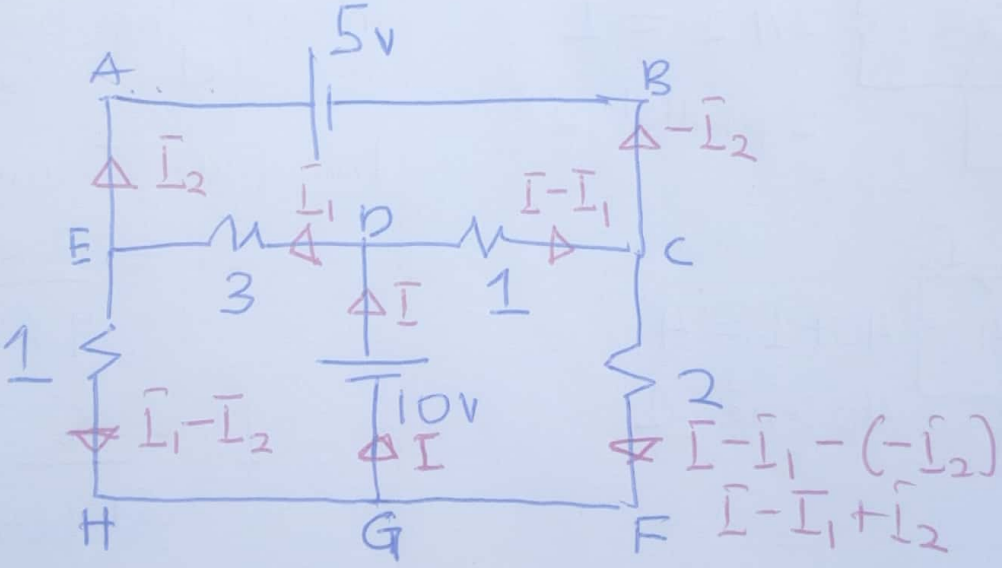
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QUESTION 2

Since The questions says using Kirchhof's laws -
—— Use KCL and KVL -

- Draw the circuit
- Assign Current on the branches.
- Write KCL and KVL equations taking loops.
- Now solve for unknows using your calculator.
- Done.
- Now use the current flowing via The 2Ω to find The power loss using $P = I^2R$,,

Solution 1



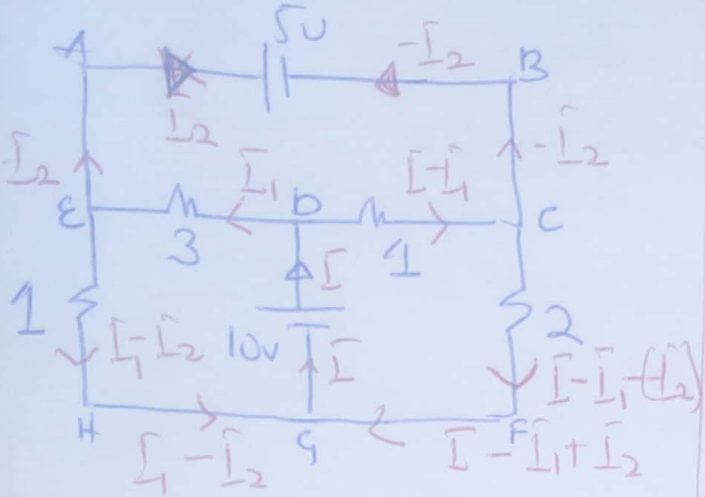
NB

① Taking a loop entering the +ve terminal of a battery gives a negative voltage and vice versa

② Taking a loop in same direction as the branch current gives rise to a −ve voltage drop. and vice versa.

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Solution 1



Using $\Sigma emf + \Sigma IR = 0$.

Taking loop ABCDEF

$$-5 + I - I_1 - 3I_1 = 0$$
$$I - 4I_1 = 5 \quad \cdots (1)$$

Taking loop DEHGD

$$10 - 3I_1 - (I_1 - I_2) = 0$$
$$-4I_1 + I_2 = -10 \quad (2)$$

Taking loop DCFGD

$$10 - (I - I_1) - 2(I - I_1 + I_2) = 0$$
$$-3I + 3I_1 - 2I_2 = -10 \quad (3)$$

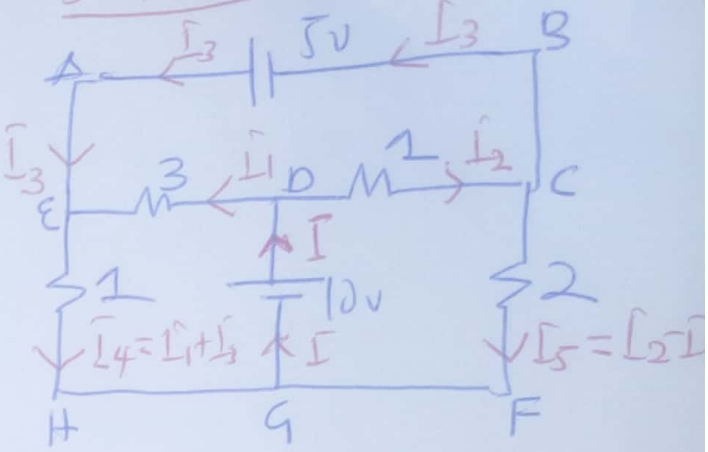
Solving for I, I_1 and I_2

$$I = 8.529\,A$$
$$I_1 = 0.8824\,A$$
$$I_2 = -6.471\,A.$$

$$I_{2n} = 8.529 - 0.8824 - 6.471$$
$$= 1.176\,A$$

$$P_2 = 1.176^2 \times 2 = \boxed{2.7640\,W}$$

Solution 2



At node E $I_4 = I_3 + I_1$

node D $I = I_1 + I_2$

C @ $I_5 = I_2 - I_3$

Taking loop ABCDEA

$$-5 + I_2 - 3I_1 = 0$$
$$-3I_1 + I_2 = 5 \quad (1)$$

Taking loop DEHGD

$$10 - 3I_1 - (I_3 + I_5) = 0$$
$$-4I_1 - I_3 = -10 \quad (2)$$

Taking loop DCFGD

$$10 - I_2 - 2(I_2 - I_3) = 0$$
$$-3I_2 + 2I_3 = -10$$

Solving simultaneously —

$$I_1 = 0.882\,A$$
$$I_2 = 7.647$$
$$I_3 = 6.471\,A$$

$$I_{2n} = I_2 - I_3 = 7.647 - 6.471$$
$$= 1.176\,A.$$

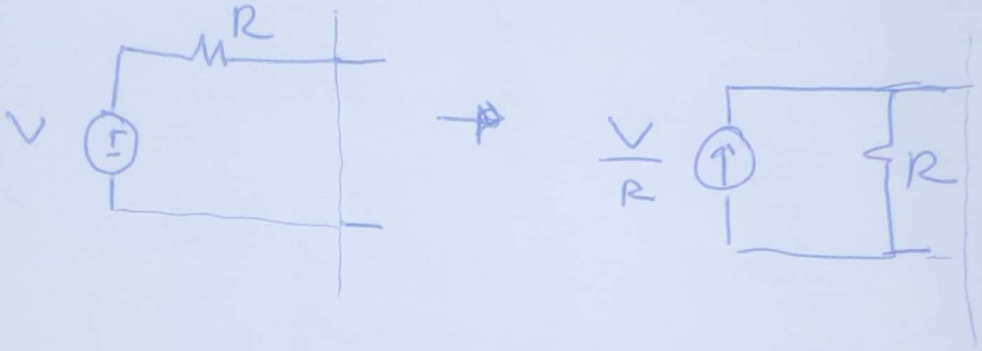
$$P_{2n} = 1.176^2 \times 2 = \boxed{2.7640\,W}$$

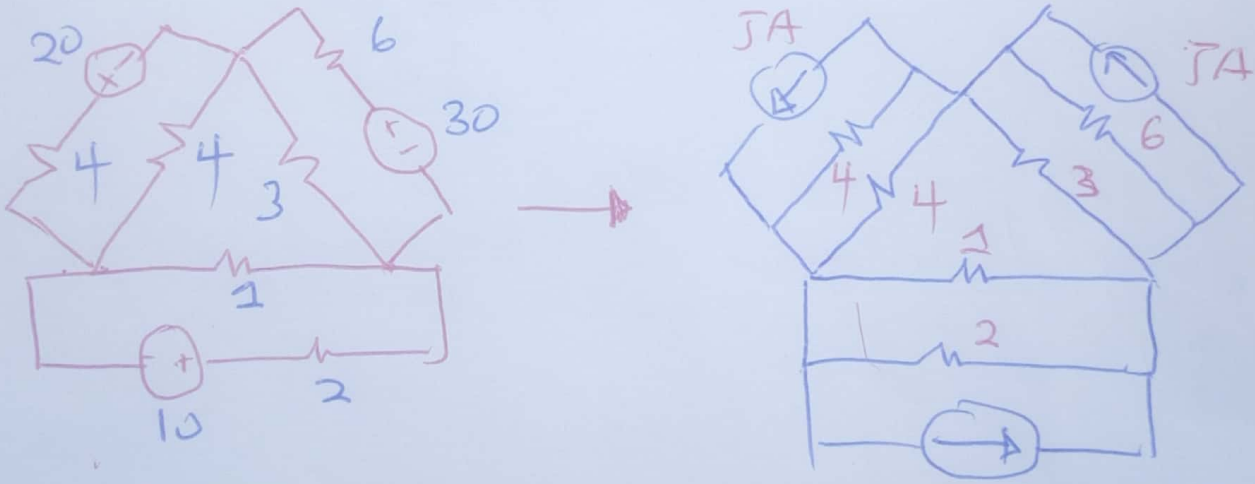
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QUESTION 3

(N3) Source transformation
- Convert voltage sources into current sources.
- Check for the flow of currents
- Combine the resistors.

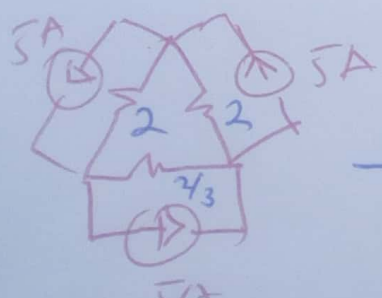
(A3) Converting Voltage source → Current Source.





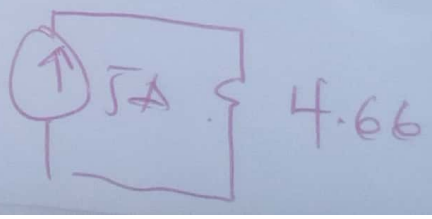
Solve the parallel combination of the resistors.

$4 \| 4 = 2$, $1 \| 2 = 2/3$, $3 \| 6 = 2$,

(A) 

→ Note that same current flows through the circuit (A)

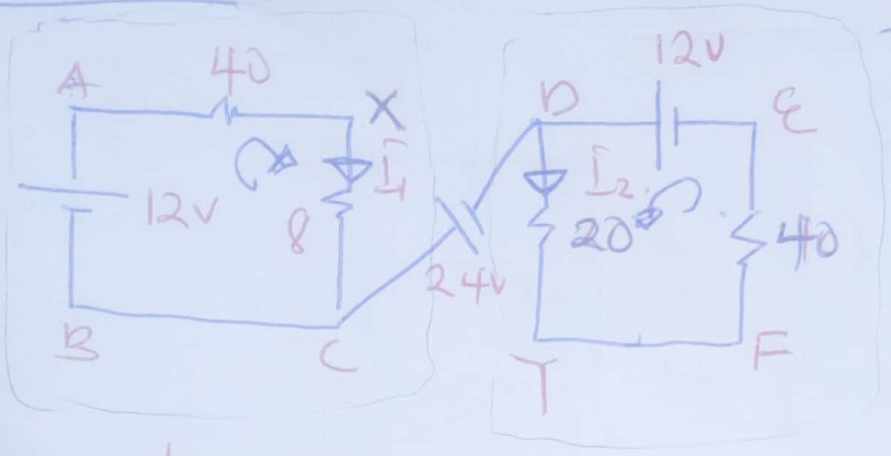
$R_T = 2 + 2 + \frac{2}{3} = 4.666.$



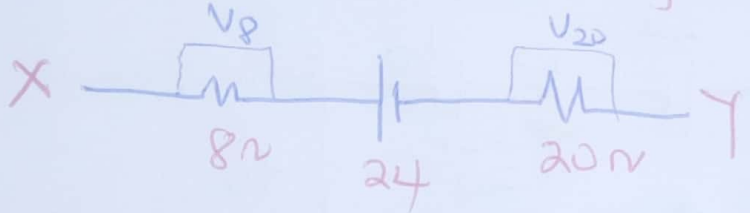
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QUESTION 4.

B ~



We are interested in the voltage across X → Y



- hence find the voltage drop across the 8Ω and 20Ω.

(NB) If current flows from circuit in the right (CB) to (A) it can't come back and vice versa hence no current flows via branch **CD**.

$$I_1 = \frac{V}{R} \quad \text{but } R = 40+8 \cdot \quad V = 12$$

$$= \frac{12}{40+8} = 0.2A.$$

$$V_8 = 8 \times 0.2 = 2V$$

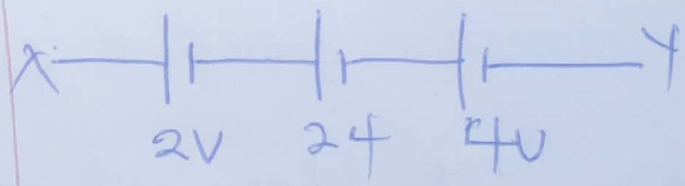
$$I_2 = \frac{V}{R} = \frac{12}{20+40} = \frac{12}{60}$$

$$= 0.25.$$

$$V_{20} = 0.25 \times 20$$

$$= 4V$$

Now we realise that the current enter via the positive terminal of the voltage drops — hence



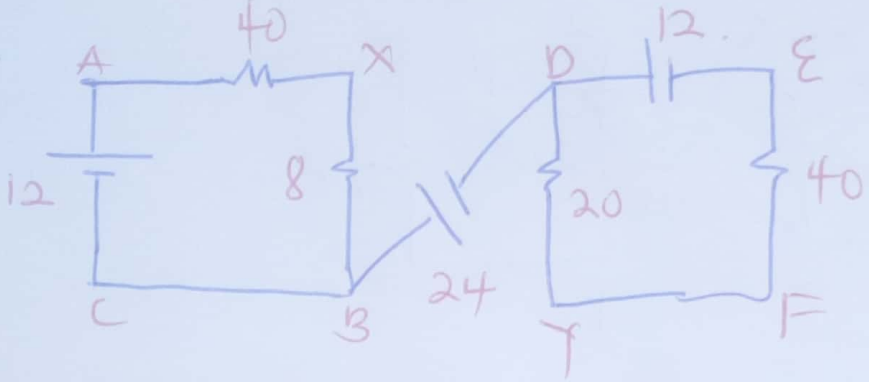
$$V_{XY} = 2 + 24 + 4V$$

$$= \boxed{30V}$$

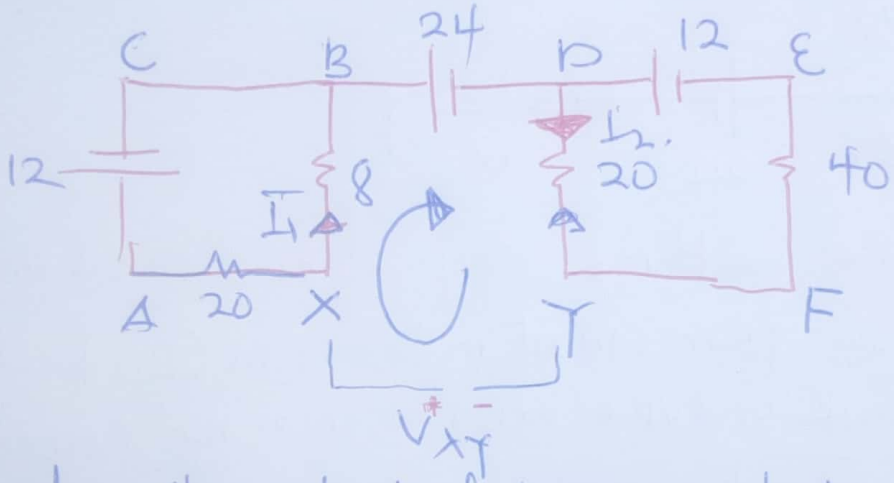
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Question 4 — Solution:

(A)



— We can redraw the circuit as.



We realize the potential difference between XY and the voltage seen at V_{XY}.

$\boxed{I_1 = 0.2}$ from previous calculation and $\boxed{I_2 = 0.25}$

Now taking loop XBDYX

$$\Sigma emf + \Sigma IR = 0.$$

$$V_{XY} - 24 - 8(0.2) - 20(0.25) = 0.$$

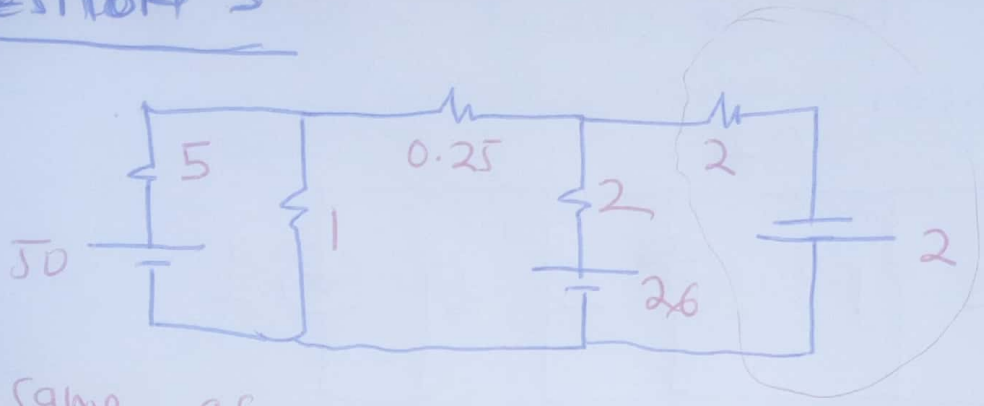
$$V_{XY} - 24 - 2 - 4 = 0.$$

$$V_{XY} - 30 = 0.$$

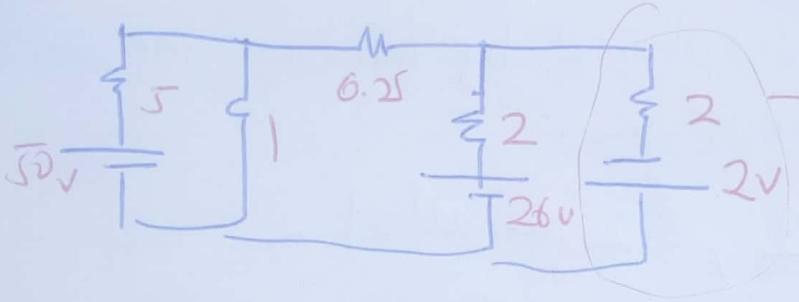
$$V_{XY} = \boxed{30\,V.}$$

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QUESTION 5



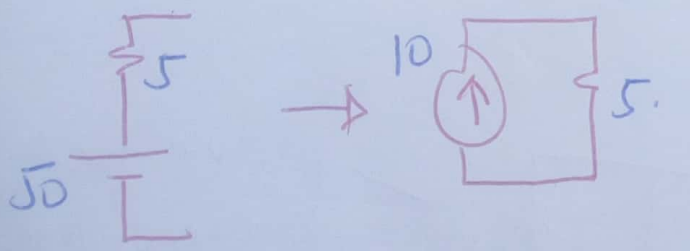
is same as.

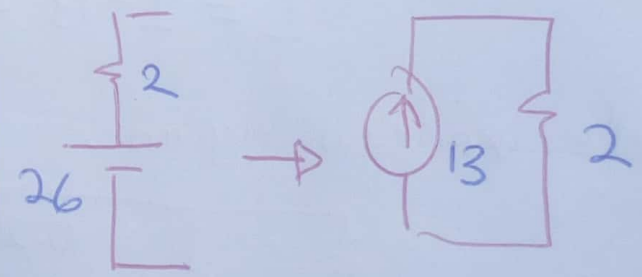


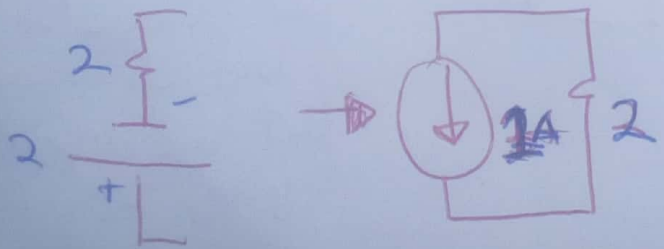
(NB) Convert all voltage sources into current sources. NB to convert voltage source into current source the V must be in series with R. after converting the I becoma parallel with R.



Solution - From the queston







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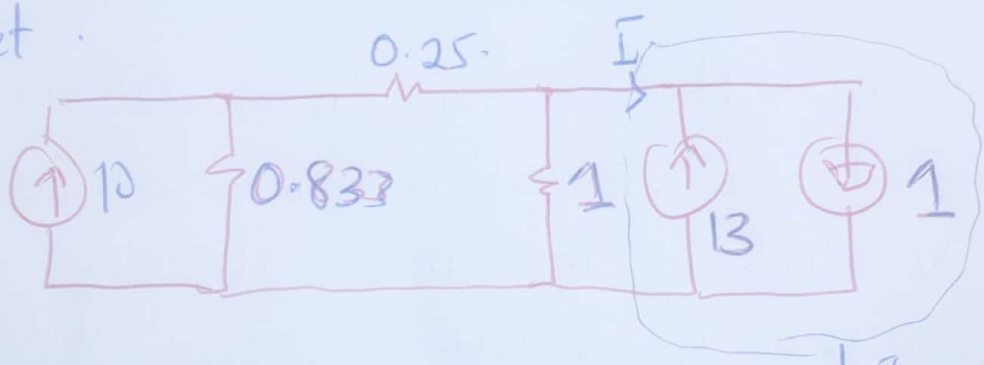
Now drawing The ~~for~~ circuit with The convected sources.



The circuit can be simplified, by combing The ~~for cut~~ Sources and the resistance.

5 parallel 1, 2 parallel 2

We get.

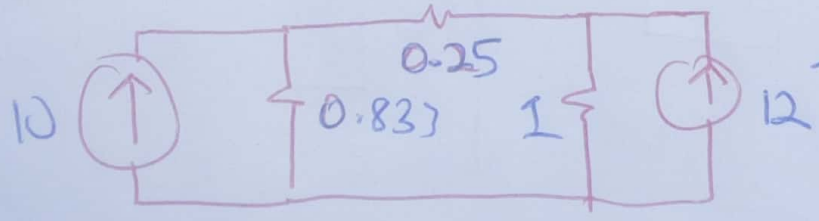


These two current will result to I

To find I, We apply KCL

$$I + 13 = 1.$$
$$I = 1 - 13 = -12A.$$

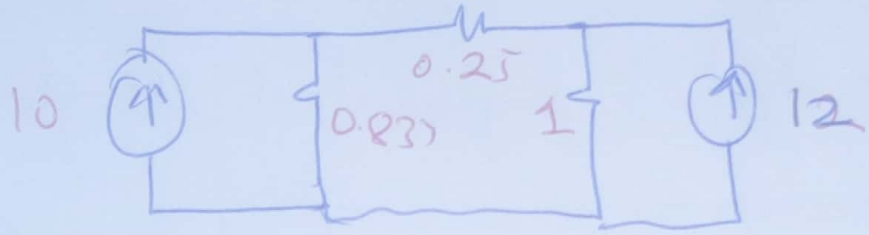
We redraw again,



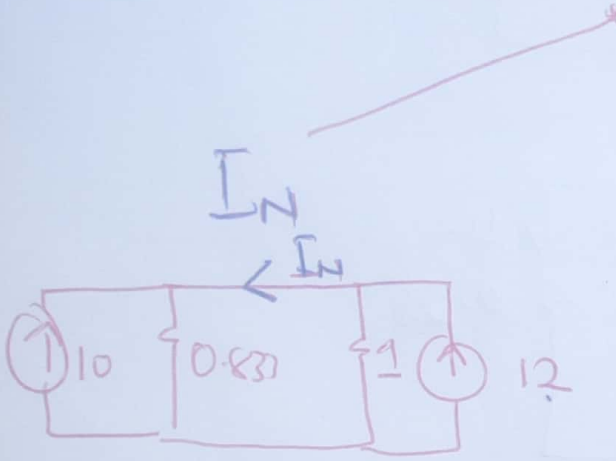
positive 12A because the current ~~was~~ per calculation flow downward but we drew it going up

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Now - We have



We now apply Nortons Theorem to find the current via 0.25Ω.



R_N

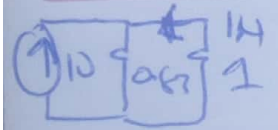
- Short 0.25 and replace it with I_N.

① We can see that because of the short circuit.

$$I_N = 12 - 10$$
$$= 2A.$$

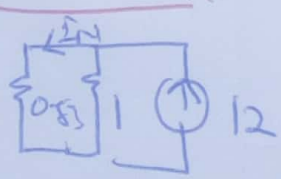
② using Supper position

10A alone



$$I_N = \frac{0.833}{1.833} \times 10$$
$$= 4.544$$

12A alone

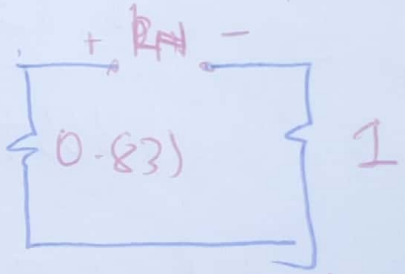


$$I_N = \frac{1}{1+0.833} \times 12$$
$$= 6.544A$$

Per current direction

$$I_N = 6.544 - 4.544 = 2A.$$

Remove all sources and ~~put~~ Replace 0.25 with R_N



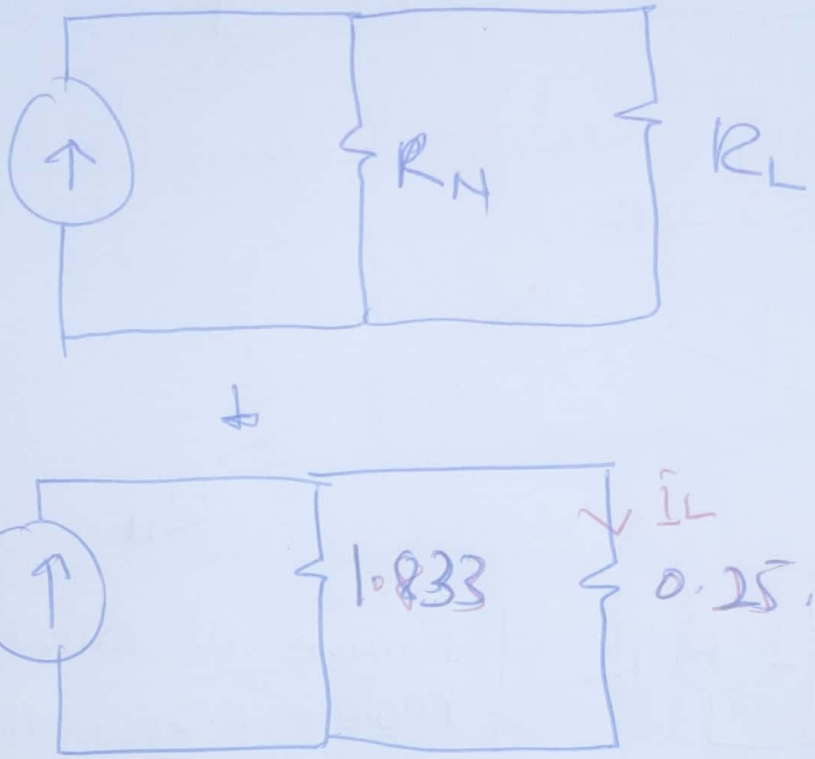
We can see clearly that 0.833Ω is in series with 1Ω.

Hence $R_N = 0.833 + 1 =$
$= 1.833\ \Omega.$

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Now That I_N and R_N is found.

Drawing The Nortons Equivalent Circuit.



$$I_L = \frac{1.833}{1.833 + 0.25} \times 2$$

$$= 1.75996\ A$$

$$= \boxed{1.76\ A}$$

Thank You

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